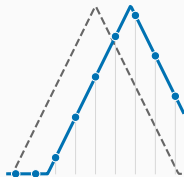


1 Complex 9-tap I/Q

prompt replica



$z_{i,k}, k = 1, \dots, 9$

0.125-chip spacing / PRN

2 Signed-delay sensor

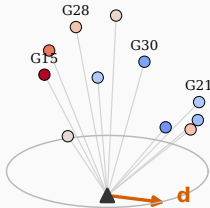
normalize + rotate

two-path dictionary

$\hat{\delta}_i, q_i$

measurement, not ML

3 PRN--LOS geometry



$$\hat{\delta}_i \approx -\mathbf{u}_i^T \mathbf{d} + c$$

4 Nested-model score

$$\mathcal{H}_0: \hat{\delta}_i = c \quad \text{SSE}_0$$

$$\mathcal{H}_1: \hat{\delta}_i = -\mathbf{u}_i^T \mathbf{d} + c \quad \text{SSE}_1$$

$$F = \frac{(1-r_G)(N-4)}{3r_G}$$

upper tail p_F

clean only $\rightarrow \eta = 0.0603$

5 Causal decision

support / rank
LOS conditioning

$p_F \leq \eta$ in $\geq 3/5$ bins

- Spoof-like alarm
- ✗ Multipath / noncoherent
- △ Gate fail: abstain